FINAL DESIGN PRESENTATION

Battery Management and Communication System

- 10s1p Li-ion instrumentation & control
- BQ76930 monitoring & STM32 firmware
- Operator dashboard
- NVIDIA Jetson AI expansion path

GROUP MEMBERS

Kaushik Vada · Christian Kim · Nick Poon · Connor Stewart

PROJECT STATE

Integrated BMS + Dashboard

Concept and Application of the Design

- Split-domain architecture for safety
- Power Domain: Battery, sensing, cooling, protection
- Host Domain: STM32 control and desktop UI

FINAL BMS ENCLOSURE RENDER (INTERACTIVE)

BATTERY PACK AND ENCLOSURE TEST HARDWARE (INTERACTIVE)

Intended Applications

- Lab battery characterization
- Embedded controls experiments
- AI battery health estimation on Jetson

What We Designed

- 10s1p Li-ion monitoring
- Current & thermal sensing
- Active cooling & balancing

Technical Principles

- BQ76930 cell supervision
- 20mΩ shunt & 10K NTC sensing
- PWM fan control

Relevant EE Subjects

- Power electronics & embedded systems
- PCB design & controls
- Hardware-software co-design

Technical Design Objectives

Translating project requirements into explicit numerical targets from the design report and firmware code.

Battery Coverage

10s1p

Monitor 10 cell voltages, 10 NTCs, pack current, and fan telemetry.

IMPLEMENTED COVERAGE

Pack Electrical Target

36 V / 4.2 Ah

Support Molicel P42A pack (151.2 Wh nominal) with cell-level visibility.

PACK SPECIFICATION

Current Resolution

0.422 mA/LSB

BQ76930 coulomb-counter + 20mΩ shunt for pack current estimation.

IMPLEMENTED CONVERSION

Safety Response Window

10-100 ms

Fast thermal/overcurrent monitoring; slower 1-10s balancing.

FIRMWARE ARCHITECTURE TARGET

Cooling Control

1 kHz PWM

0-100% duty, 2°C hysteresis, 4-sample RPM avg, 2s stall detection.

BENCH-IMPLEMENTED

Host Interface

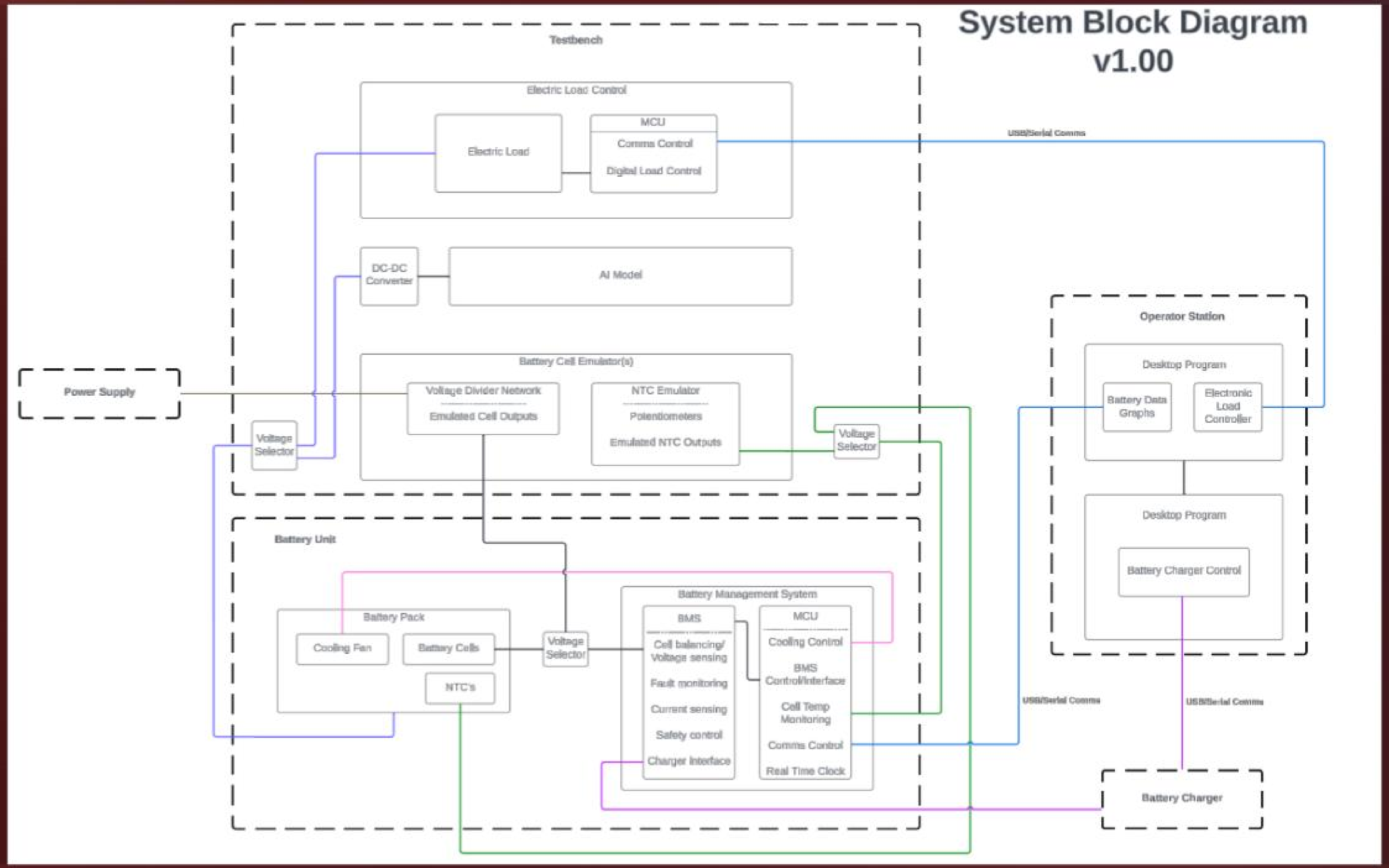
115200 baud

USB CDC telemetry, QWebChannel command delivery, 50ms polling.

INTEGRATED PATH

Final High-Level Design

- System partitioned into battery unit, control electronics, and operator station
- Isolates high-energy pack from USB host interface



SYSTEM BLOCK DIAGRAM SHOWING BATTERY UNIT, MONITORING ELECTRONICS, BENCH SUPPORT, AND OPERATOR STATION

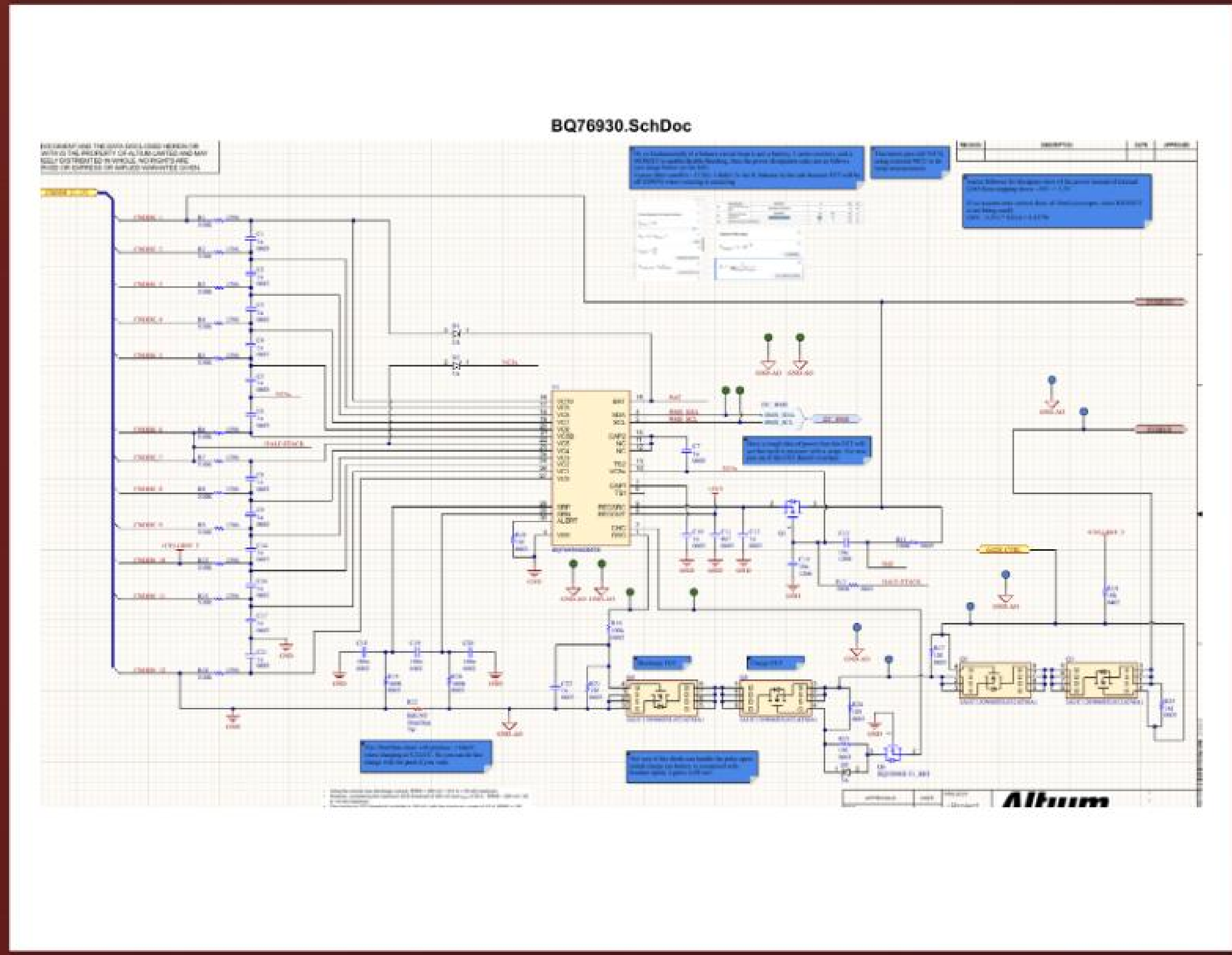
Battery Domain
10s1p pack, thermistors, fan, balancing, protection hardware.

Control Domain
STM32F303, ADCs, BQ76930 I2C, PWM, USB CDC.

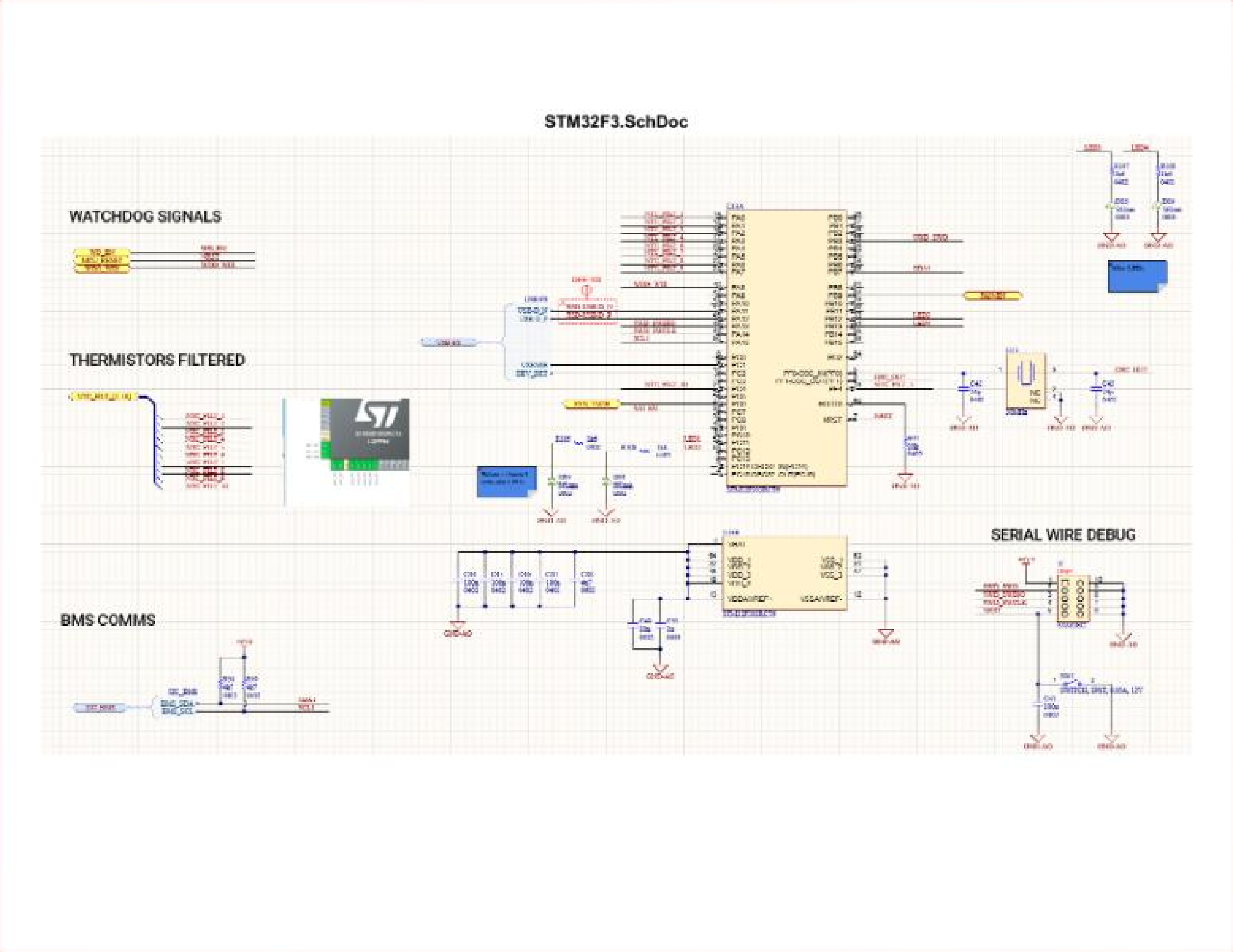
Host Domain
PyQt6 dashboard, command GUI, Jetson AI pipeline.

Final Low-Level Design: Electrical Implementation

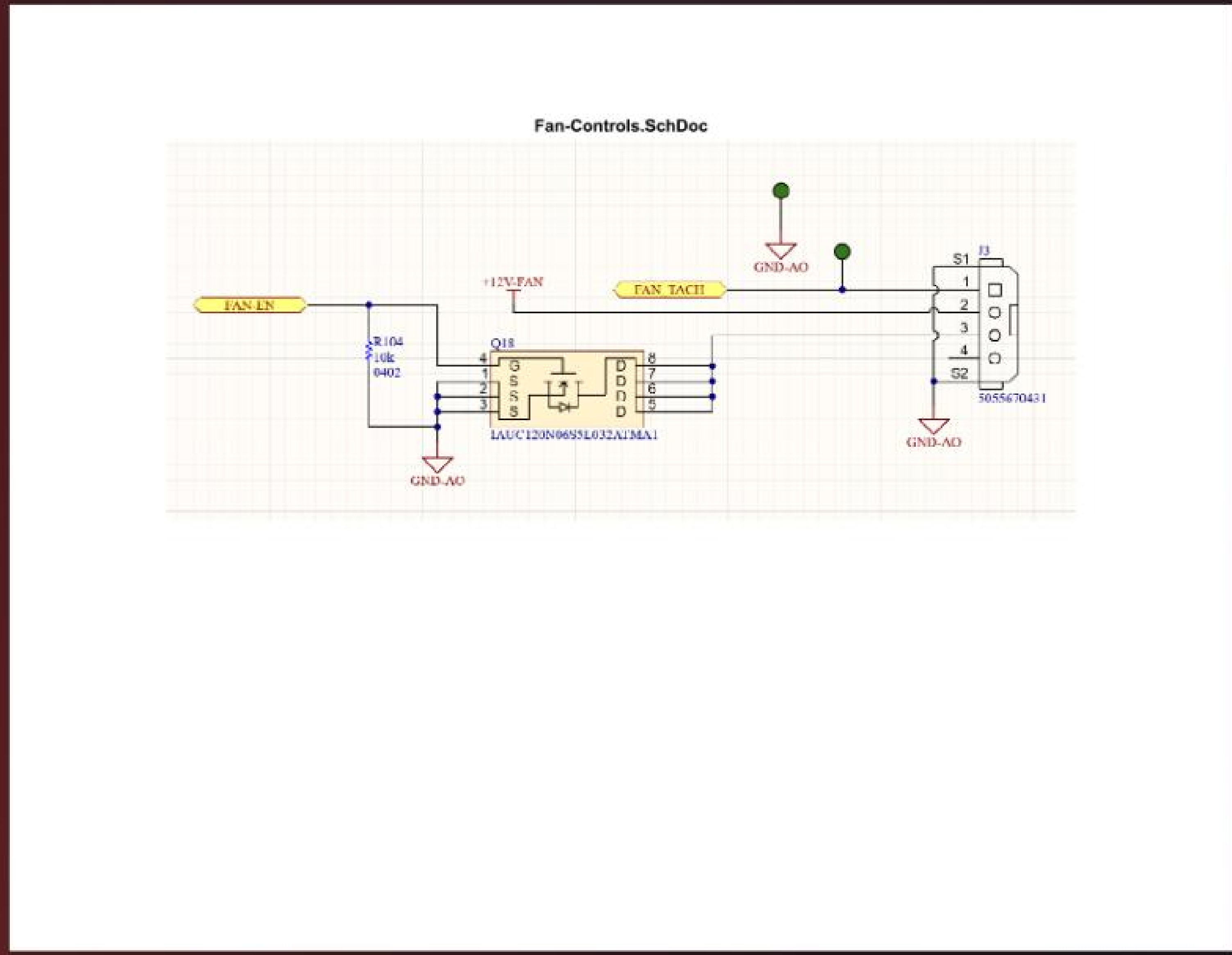
Core low-level implementation across three main schematic pages.



BQ76930 FRONT-END: CELL TAPS, SHUNT CURRENT PATH, CHARGE/DISCHARGE FET CONTROL



STM32F303RC: THERMISTOR INPUTS, BMS COMMS, WATCHDOG, LEDS, USB AND DEBUG I/O



FAN CONTROL SHEET: PWM ENABLE, TACH FEEDBACK, CONNECTOR ROUTING, AND POWER INTERFACE

Final Low-Level Design: Program Flow-Charts

Implementation flow logic synthesized directly from firmware and dashboard operations.

Firmware Runtime Flow

1. Boot and bring-up
Init clocks, ADCs, I2C, timers, USB, GPIO.

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2. BMS interface detection
Probe BQ76930, auto-detect CRC.

↓

3. Sensor acquisition
Sample 10 NTCs, cell voltage/current, apply Beta-model math.

Desktop Dashboard Flow

1. Launcher startup
Start PyQt6 WebEngine, init bridge, load UI.

↓

2. Serial connection layer
Auto-detect USB CDC at 115200 baud.

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Technical Challenges

Key integration challenges across sensing, firmware, comms, and packaging.

SIGNAL STABILITY

High-impedance thermistor readings were noisy

Problem: Noisy high-impedance NTC values at fast ADCs.
Solution: Increased sampling time, used Beta-model math.
Workaround: Next Step: Full reference thermometer calibration.

FIRMWARE INTEGRATION

PWM fan control required missing HAL timer pieces

Problem: Missing HAL TIM drivers for PWM.
Solution: Added TIM drivers, tuned to 1 kHz PWM & 2°C hysteresis.
Workaround: RPM falls back to duty cycle on timeout.

PROTOCOL ROBUSTNESS

Mixed serial formats broke clean frame reconstruction

Problem: Mixed serial formats caused incomplete reads.
Solution: Regex parser, frame assembly, 0.3s timeout.
Workaround: Fallback: GUI simulation mode.

MECHANICAL PACKAGING

Compact enclosure packaging fought airflow and fit

Problem: Compact footprint impacted tolerances & airflow.
Solution: Iterative CAD revisions for fit & airflow.
Workaround: Next Step: Production-ready harness & enclosure refinement.

Major Components of the Design and Implementation

Primary implementation ownership per subsystem.

Battery Pack and Harness

- 10s1p cell holder
- Enclosure fit-up
- Harness routing

Owners: Nick, Christian

BMS Hardware Stack

- BMS schematic/layout
- BQ76930 monitoring
- Protection interfaces

Owners: Connor, Nick

Embedded Control Firmware

- STM32F303RC bring-up
- ADC & sensing integration
- Fan & USB CDC control

Owners: Kaushik, Connor

Operator Dashboard

- PyQt6 launcher & bridge
- Command-surface GUI
- Jetson data path

Owners: Kaushik, Christian

Who Contributed to What

Subsystem leadership across the team.

KAUSHIK

Dashboard and System Integration

- Dashboard and GUI design/implementation
- CAD and enclosure refinement
- Firmware integration and telemetry path work
- NVIDIA Jetson AI model pipeline direction

CHRISTIAN

E-Load and Bench Validation

- E-Load development
- Testing and debugging support
- NVIDIA Jetson AI model contribution
- Battery charger and pack/enclosure harness setup

NICK

PCB and Validation Hardware

- BMS layout and PCB design
- E-Load validation
- BMS validation
- Battery charger and battery-cell harness engineering

CONNOR

Architecture and Schematic Direction

- Overall planning and system planning
- BMS schematic development
- Emulation board design
- Firmware assistance and architecture support

Design Considerations

- System shaped by schedule, thermal, packaging limits
- Uses robust, reliable communication standards

Time and Integration Risk

- Tolerant serial framing
- Simulation mode
- Staged bring-up

Power, Thermal, and Safety Tradeoffs

- 1 kHz PWM fan path
- Breakpoint auto-curve
- 2°C hysteresis
- Clear fault observability

Size, Weight, and Cost

- Compact enclosure with optimized airflow
- Reused STM32Cube/HAL & USB CDC
- Iterative 3D-printed revisions

Industry Standards and Interfaces Used

USB 2.0 CDC	Host telemetry and command interface
I2C	BQ76930 communication and RTC-style peripheral planning
ADC + NTC sensing	10K Beta-model thermistor measurement path
PWM + Tach capture	Fan duty-cycle actuation and RPM feedback
JSON / tokenized telemetry	Structured frontend state reconstruction during bring-up
QWebChannel	Browser-to-Python bridge inside the packaged desktop app

Test Report

Validated behaviors based directly on repo documentation and implementation.

TEST ITEM	ORIGINAL OBJECTIVE	OBSERVED / IMPLEMENTED RESULT	STATUS
Cell and thermal visibility	Show all 10 cells and 10 thermistor channels	10 cell voltages plus thermistor mapping to the GUI are implemented	MET
USB host link	Reliable host communication at 115200 baud	USB CDC plus GUI serial connect flow defaults to 115200	MET
Fan control loop	Auto/manual cooling with measurable RPM	1 kHz PWM, 0-100% duty, 2 deg C hysteresis, 2 s RPM timeout, 4-sample average	MET
Current sensing	Convert BQ current registers into pack current	20 mOhm shunt conversion implemented at 0.422 mA/LSB	MET
Parser resilience	Handle mixed telemetry formats without dropping frames	Regex/token parser, JSON handling, and 0.30 s multiline finalize path are implemented	MET
Calibration completeness	Close the loop with calibrated live hardware data	Integration and calibration are identified next steps in the repo	PARTIAL

Validation Strategy

- Firmware checks
- Serial reconstruction tests
- GUI verification
- Staged bench testing

How Close to Spec?

- Implemented: Comms, telemetry, fan control, monitoring.
- Remaining gap: Calibration & live scale testing.

Key Quantitative Anchors

- 10 cells, 10 NTCs
- 115200 baud, 1 kHz PWM
- 2°C hysteresis, 0.422 mA/LSB
- 2s CMD timeout, 50ms fallback

SLIDE 12

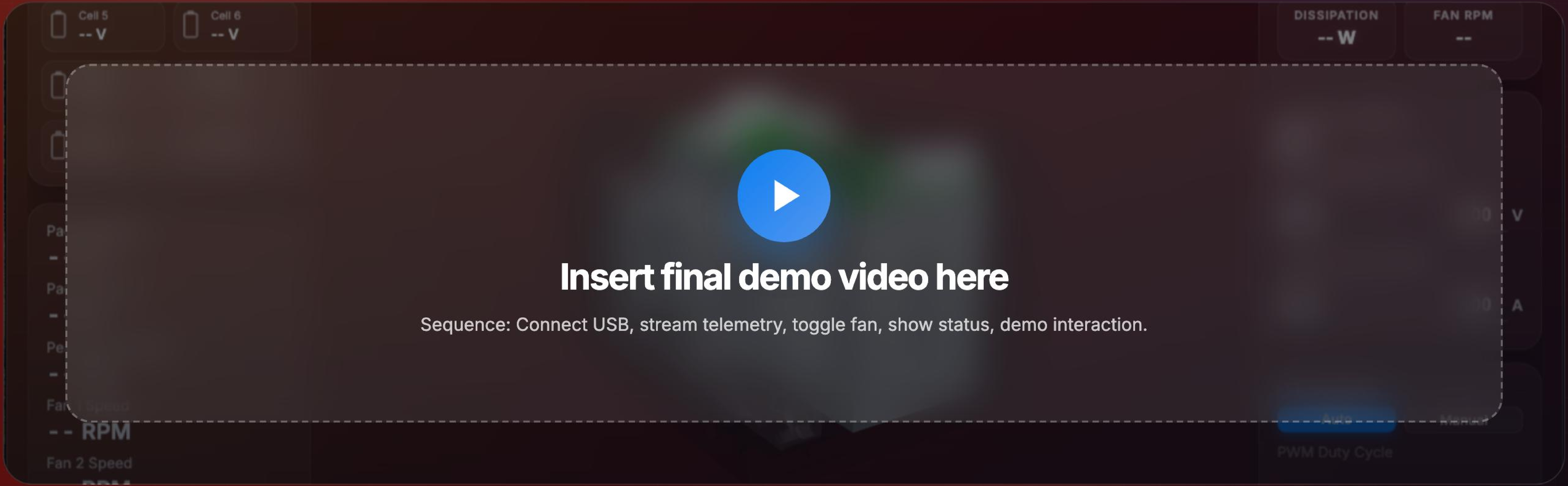
DEMO

Video placeholder for the final PowerPoint

Video

Demo Video Placeholder

Reserved space for the recorded demo clip.



Insert final demo video here

Sequence: Connect USB, stream telemetry, toggle fan, show status, demo interaction.

Scene 1
Dashboard boot and serial connection

Scene 2
Live cell, current, and temperature telemetry

Scene 3
Fan auto/manual control and status updates

Scene 4
E-Load or charger interaction and overall system response

Summary

Delivered cross-domain engineering system with a path toward AI analytics.

What Was Achieved

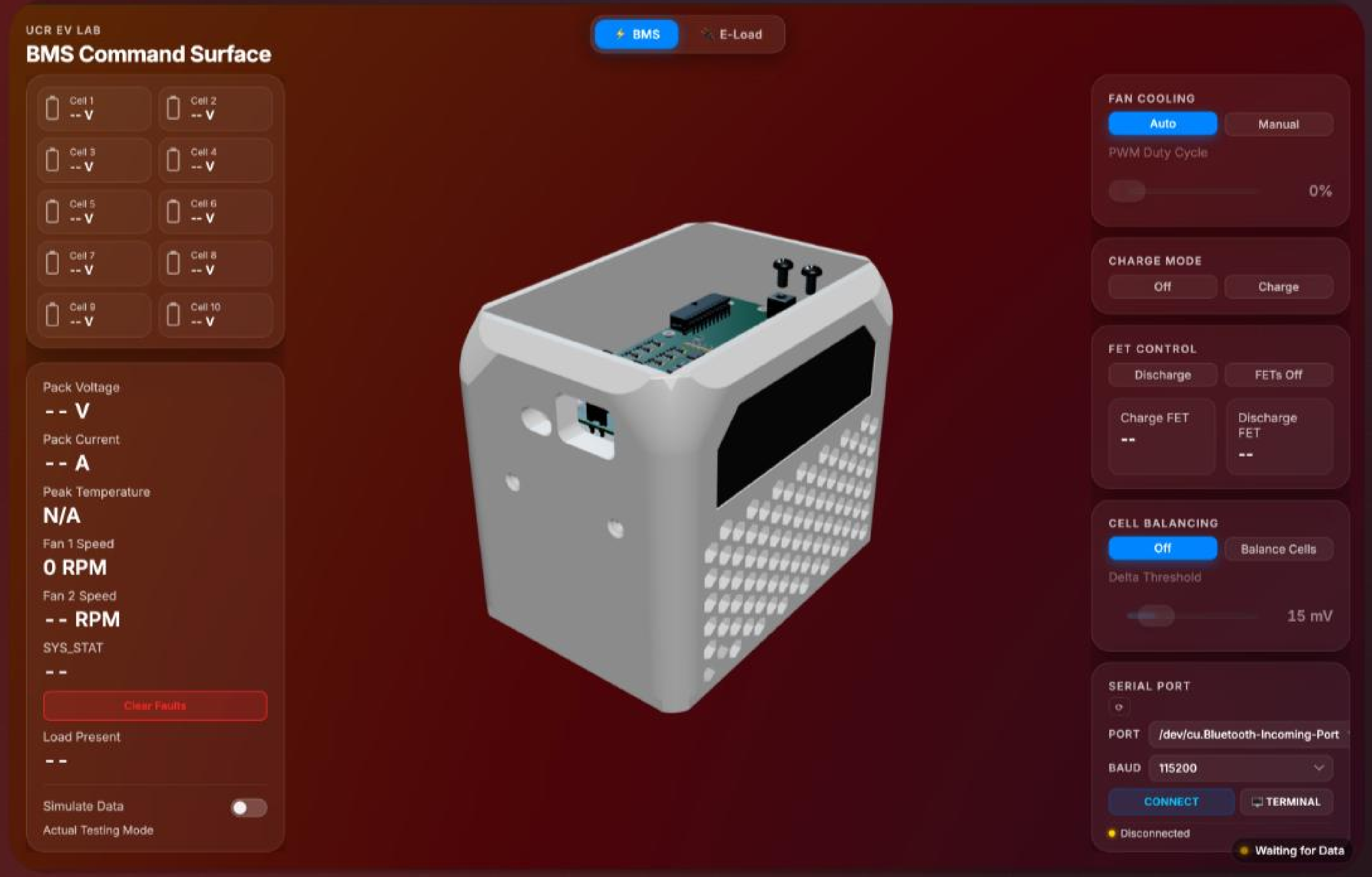
- BQ76930 sensing & STM32 control
- USB telemetry & premium dashboard
- Validated architecture

What Is Strongest

- Cell visibility & current measurement
- Thermal control
- Parser robustness & GUI usability

What Comes Next

- Final calibration & closer charger validation
- Fault handling & trending
- Jetson-based state estimation



FINAL COMMAND SURFACE PRESERVING THE PROJECT VISUAL LANGUAGE

Closing Takeaway

A modular, safe, expandable battery platform ready for analytics and validation.

Sources used in this deck: final report, weekly reports, firmware plans, dashboard source, schematic PDFs, and project assets from the EE175 workspace.